

BOGUCHANY, Russian Federation

WMO: 292820

Lat: **58.3838N**Lon: **97.4507E**Elev: **438**StdP: **14.46**Time Zone: **7.00 (E07)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-44.5	-39.0	-47.7	0.3	-43.6	-42.7	0.5	-38.5	14.9	10.4	13.2	9.7	0.3	260	0.601

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.0	85.2	65.3	81.9	64.0	78.3	62.4	68.1	79.5	66.2	77.4	64.5	75.1	4.5	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
64.2	91.5	72.7	62.1	84.9	70.6	60.3	79.5	69.3	32.8	79.7	31.3	77.3	29.9	75.0	74.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 13.6	11.8	10.3	-44.6	91.6	6.7	2.7	-49.4	93.5	-53.3	95.1	-57.1	96.6	-61.9	98.6		
(5)			-44.3	71.0	6.1	2.0	-48.7	72.4	-52.2	73.6	-55.7	74.7	-60.1	76.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	30.2	-9.5	-2.2	16.8	34.4	47.2	62.0	66.2	60.5	47.0	32.3	10.5	-5.2		
	DBStd	29.00	18.28	15.97	13.62	8.63	8.70	7.13	5.56	6.25	7.59	9.66	16.87	18.68		
	HDD50	8564	1845	1460	1029	471	162	5	0	4	145	549	1184	1710		
	HDD65	12874	2310	1880	1494	918	553	137	53	166	541	1013	1634	2175		
	CDD50	1332	0	0	0	2	76	365	503	330	55	1	0	0		
	CDD65	167	0	0	0	0	2	48	91	26	0	0	0	0		
	CDH74	1826	0	0	0	0	66	620	855	274	11	0	0	0		
	CDH80	484	0	0	0	0	12	176	238	57	1	0	0	0		
	WSAvg	3.8	3.0	3.2	3.9	4.9	4.8	3.7	3.2	3.2	3.8	4.7	4.5	3.3		
	PrecAvg	13.5	0.6	0.5	0.4	0.7	1.3	1.7	2.0	2.0	1.7	1.0	1.0	0.8		
(15) Precipitation	PrecMax	19.6	1.4	0.9	1.0	1.7	3.1	5.3	6.0	4.8	4.1	2.6	1.6	1.5		
	PrecMin	8.7	0.2	0.2	0.0	0.1	0.2	0.0	0.2	0.2	0.5	0.2	0.3	0.2		
	PrecStd	2.7	0.3	0.2	0.2	0.3	0.6	1.1	1.3	0.9	0.8	0.5	0.3	0.3		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	31.4	34.9	48.9	64.5	81.0	89.4	90.0	86.2	74.9	56.5	44.5	33.1		
		MCWB	28.4	30.2	38.5	46.9	57.3	64.6	68.3	66.8	59.0	46.3	39.8	30.2		
	2%	DB	26.8	31.2	42.5	58.2	74.3	85.0	85.6	80.8	67.9	51.5	39.2	29.5		
		MCWB	24.5	27.7	34.7	44.6	54.9	63.5	66.5	64.3	55.7	44.9	35.5	27.2		
	5%	DB	22.1	27.4	38.7	52.8	68.9	81.2	82.5	76.7	62.8	48.0	36.4	25.7		
		MCWB	20.0	24.4	32.5	42.0	52.7	62.1	65.6	62.5	53.6	42.8	33.4	23.7		
	10%	DB	16.9	21.8	35.5	48.0	63.9	77.1	78.9	72.8	58.5	44.7	33.1	21.2		
		MCWB	15.3	19.6	30.5	39.2	50.4	61.0	64.0	60.7	51.7	40.5	30.7	19.4		
	0.4%	WB	28.6	31.0	39.2	49.4	60.0	68.7	71.7	68.8	61.1	48.7	40.6	30.3		
		MCDB	30.9	34.6	46.7	60.3	74.5	80.7	84.8	81.0	70.6	53.0	44.1	32.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	24.6	28.0	35.5	45.5	56.9	66.1	69.1	66.0	57.4	46.0	36.1	27.3		
		MCDB	26.8	31.1	41.2	56.3	70.8	78.1	79.9	77.9	65.3	50.3	38.8	29.4		
	5%	WB	20.0	24.4	33.1	42.7	54.1	64.4	67.4	63.9	54.9	43.2	33.4	23.8		
		MCDB	22.0	27.5	38.4	51.7	66.7	77.2	78.3	73.4	60.9	47.5	36.1	25.6		
	10%	WB	15.4	19.5	30.4	39.7	51.3	62.5	65.6	62.0	52.5	40.8	30.6	19.4		
		MCDB	17.0	21.9	35.5	47.4	62.5	74.2	75.9	70.3	57.9	44.5	33.2	21.3		
	5% DB	MDBR	14.1	19.0	22.3	18.2	20.7	21.4	20.0	18.4	15.3	11.2	11.2	13.2		
		MCDBR	14.4	17.8	21.8	26.5	29.9	29.8	26.3	25.9	23.9	16.9	10.3	14.2		
		MCWBR	13.2	15.7	15.6	15.1	14.1	11.4	10.0	11.1	12.0	11.2	8.7	13.2		
		MCWBR	14.6	17.5	20.1	23.9	27.1	24.7	21.4	22.3	19.8	15.1	10.2	14.8		
(40) Clear-Sky Solar Irradiance	taub		0.281	0.308	0.340	0.373	0.393	0.431	0.456	0.418	0.361	0.336	0.291	0.264		
		taud	1.981	2.006	2.010	2.131	2.267	2.230	2.173	2.249	2.377	2.260	2.040	1.943		
	Ebn at Noon		159	217	247	262	266	256	245	246	245	210	153	122		
		Edh at Noon	18	29	39	41	38	41	42	36	27	23	17	14		
(44) All-Sky Solar Radiation	RadAvg		192	494	921	1404	1616	1865	1716	1349	804	384	189	109		
	RadStd		24	41	65	78	113	161	113	89	113	35	29	20		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (1 neighbor)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page